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**CS 294 (Special Topics) Dynamical Systems**

Assumptions: a first course in differential equations (roughly equivalent to CS 130);
ideally, a first course in numerical methods (roughly equivalent to CS 131).

Credit: 3 u

We will study the dynamics of various systems, deterministic and stochastic, paying special attention to non-autonomous dynamical systems. We intend to solve, computationally, problems in Science and Engineering.

We expect class members to be able to work with available software, or could find suitable software on their own, modifying them as they have to.

The elements of this course will be:

a) Problem sets (group work), which will require hand-calculations and analysis 20 %
b) Reports/ Machine problems (individual). 20 %. You will be asked to present to the class, round-robin order.
c) Project/ Presentation at the end of the term, where you will discuss the dynamics of a system you are interested in. 10 %
d) Two written exams (around midterm and finals), 50 %

**Recommended Texts:**

"Nonlinear Dynamics and Chaos", by S.H. Strogatz. 1994. Perseus Books. Not in library. May photocopy mine.

"Dynamics Systems: by H.V. Vu and R.S. Esfandiari. 1997 McGraw-Hill.

"Dynamical Systems and Automatic Control: by J.L. Martins de Carvalho. 1993, Prentice Hall.

**Important Dates:**

Midsem: Wed, Oct. 5

Last Day of dropping: Thu, Oct. 27

On-line Group Discussions: These are due in a single PDF file at 11:59 pm on due date.

Teams share one grade (75%) plus member evaluation (25 %). Peer evaluations require you give your team members a number from 1 to 10, 10 being the best. Ten points off for everyday late.

Reports: Ten points off for everyday late.